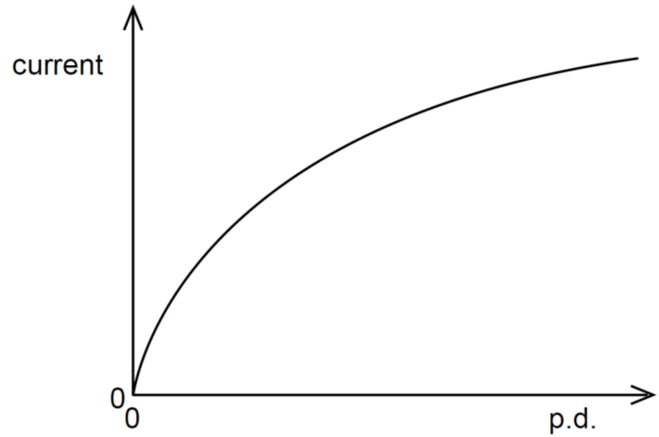
21	The graph shows how the current through a lamp filament varies with the potential difference (p.d.) across it.



Which statement explains the shape of this graph?

A	As the filament temperature rises, electrons can pass more easily through the filament.
B	It takes time for the filament to reach its working temperature.
C	The power output of the filament is proportional to the square of the current through it.
D	The resistance of the filament increases with a rise in temperature.

Ans: D – resistance is the ratio of p.d. over current, and it can be seen from this graph that it is increasing as the p.d., thus temperature, increases.